

Comparative Analysis of Pre-GAP and Post-GAP Knowledge Distillation

From Pretrained Image Classifiers to Lightweight ConvNet Students

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Outline

- 1 Introduction and Motivation
- 2 Methodology
- 3 Experimental Setup
- 4 Results
- 5 Conclusion

SECTION 1

1. Context

- Deep learning models can achieve strong performance in visual recognition tasks.
- Larger models often require more computational resources for training and inference.
- The final project studies this trade-off through controlled experiments and quantitative evaluation.

1. Motivation

Main Motivation

Understand how different deep learning training strategies affect accuracy, generalization, and computational cost.

- Compare baseline and improved training configurations.
- Analyze model behavior across datasets.
- Identify practical limitations and future improvements.

SECTION 2

2. Method Overview

- Define the classification task and selected datasets.
- Train reference models under controlled experimental settings.
- Evaluate student or compact models using the same metrics.
- Compare results across datasets, architectures, and training strategies.

2. Pipeline



Goal

Produce a reproducible comparison that supports the final experimental conclusions.

SECTION 3

3. Datasets

Dataset	Classes	Train	Test
Dataset 1	TBD	TBD	TBD
Dataset 2	TBD	TBD	TBD
Dataset 3	TBD	TBD	TBD

3. Training Configuration

Parameter	Value
Optimizer	TBD
Learning Rate	TBD
Batch Size	TBD
Epochs	TBD
Loss Function	TBD
Hardware	TBD

3. Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Model size
- Computational cost

SECTION 4

4. Main Results

Model	Accuracy	Precision	Recall	F1
Baseline	TBD	TBD	TBD	TBD
Proposed	TBD	TBD	TBD	TBD
Reference	TBD	TBD	TBD	TBD

4. Result Analysis

- Which configuration performed best?
- Which dataset was more challenging?
- Did the proposed method improve over the baseline?
- What trade-offs appeared between performance and cost?

SECTION 5

5. Conclusions

- The project evaluated deep learning models under a controlled experimental setup.
- The experiments highlight the strengths and limitations of each configuration.
- The final results support the main observations and future directions.

5. Future Work

- Test additional architectures.
- Improve hyperparameter search.
- Add stronger baselines.
- Analyze model efficiency in more detail.

Questions?

Thanks!